

A Phenomenological Model for Joint-Bath Quantum Decoherence in Molecule Interferometry: Path Information and Thermal Dephasing with a Bilinear Cross Term (DTQEM v18.0-C)

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We present a phenomenological model for quantum decoherence in systems simultaneously subject to path-information extraction and thermal dephasing. The coherence function $C = C_0 \exp(-a I_{\text{path}} - b \Delta T/T_{\text{ref}} - c I_{\text{path}} \Delta T/T_{\text{ref}})$ extends the independent-bath baseline (v17.0-C, $c = 0$) by a bilinear cross-coupling term c , capturing the synergistic action of correlated environmental modes. Calibrated at $T_{\text{ref}} = 300$ K, the conservative ($c = 0$) limit achieves $R^2 = 0.9679$ and LOOCV $R^2 = 0.9356$; the full model reaches $R^2 = 0.9982$ (LOOCV = 0.9814) on the same synthetic dataset. Re-fitted to the eight-point dataset of Hornberger *et al.* (2003), the functional form yields $R^2 = 0.936$, demonstrating compatibility with molecule-interferometry data. A second-order cumulant analysis suggests strong inter-bath correlations, and an exploratory numerical scan reveals a thermal crossover at $T^* \approx 3.82$ K, flagged as a preliminary observation requiring low-temperature experimental confirmation. All code is openly available. A companion paper (Paper II) will provide the full microscopic derivation.

I. INTRODUCTION

Quantum decoherence—the irreversible loss of phase coherence in quantum superpositions—limits every practical quantum technology. Two physically distinct channels often coexist: (i) which-path information leakage, reducing fringe visibility via the Englert–Greenberger–Yasin (EGY) relation; and (ii) thermal dephasing from stochastic phase kicks.

In the simplest treatment these channels add independently:

$$C_{\text{v17}} = C_0 \exp\left(-a I_{\text{path}} - b \frac{\Delta T}{T_{\text{ref}}}\right), \quad (1)$$

where $I_{\text{path}} \in [0, 1]$ is the EGY distinguishability. However, in molecular and solid-state platforms the same phonon bath often mediates both channels, producing a cross term that Eq. (1) cannot capture. We formulate a minimal extension—DTQEM v18.0-C—adding a bilinear coupling $c I_{\text{path}} \Delta T/T_{\text{ref}}$, calibrate it, and test it against independent data. This paper focuses on the phenomenological model; a companion paper (Paper II) will provide the full microscopic derivation from the Feynman–Vernon influence functional.

II. MODEL DEFINITION

A. Core equation

$$C(I_{\text{path}}, T) = C_0 \exp\left(-a I_{\text{path}} - b \frac{\Delta T}{T_{\text{ref}}} - c I_{\text{path}} \frac{\Delta T}{T_{\text{ref}}}\right), \quad (2)$$

with $\Delta T = T - T_{\text{ref}}$, $T_{\text{ref}} = 300$ K fixed, and four free parameters $\{C_0, a, b, c\}$. Setting $c = 0$ recovers Eq. (1).

B. Physical interpretation

- $a I_{\text{path}}$: EGY path-dephasing, consistent with Caldeira–Leggett weak coupling.
- $b \Delta T/T_{\text{ref}}$: linear thermal dephasing from the high- T Ohmic expansion $\coth(\hbar\omega/2k_B T) \approx 2k_B T/\hbar\omega$.
- $c I_{\text{path}} \Delta T/T_{\text{ref}}$: phenomenological cross-coupling for shared bath modes. Motivated by physical boundary conditions and a second-order cumulant argument (Appendix A); treated as a fit parameter.

III. CALIBRATION

A. Dataset and procedure

Calibration used a synthetic grid $I_{\text{path}} \in [0, 1]$, $T \in [300, 550]$ K generated by the `dtqem_joint_bath_v18.py` simulator. Levenberg–Marquardt nonlinear least squares with leave-one-out cross-validation (LOOCV) was applied.

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B. Calibrated parameters

TABLE I. Calibrated parameters (DTQEM v18.0-C). Uncertainties: 1σ from the covariance matrix.

Parameter	Value	1σ	Role
C_0	0.3675	± 0.0052	Baseline coherence
a	1.6968	± 0.0245	Path-dephasing rate
b	0.8055	± 0.0148	Thermal-dephasing rate
c	0.5000	± 0.0201	Cross-coupling (empirical)
T_{ref}	300 K	fixed	Reference T

C. Performance metrics

TABLE II. Goodness-of-fit. LOOCV R^2 is the primary generalisation metric.

Model	R^2	LOOCV R^2	RMSE
v17.0-C ($c = 0$)	0.9679	0.9356	0.0202
v18.0-C (full)	0.9982	0.9814	0.0045

Note. Metrics computed on synthetic dataset generated from the same functional form; LOOCV assesses overfitting, not experimental generalisation.

IV. VALIDATION ON HORNBERGER (2003) DATA

Hornberger *et al.* reported C_{70} fullerene fringe visibility vs. collision rate. We extracted the eight published points, mapped collision rate to I_{path} via an auxiliary calibration, and re-fitted all four parameters independently. The re-fitted model achieves $R^2 = 0.936$.

The agreement demonstrates that Eq. (2) is *compatible* with independent molecule-interferometry data. It does not prove universality of the calibrated c . Statistical power of this $N = 8$ test is quantified in Appendix B.

V. EXPLORATORY TEMPERATURE SCAN: T^*

We define the effective thermal coefficient

$$b_{\text{eff}}(T) \equiv - \left. \frac{\partial \ln C}{\partial (\Delta T / T_{\text{ref}})} \right|_{I_{\text{path}}} \quad (3)$$

and evaluate it by numerically integrating the spin-boson dephasing functional

$$\Phi(t) = \int_0^\infty \frac{J(\omega)}{\omega^2} \coth\left(\frac{\hbar\omega}{2k_B T}\right) (1 - \cos \omega t) d\omega \quad (4)$$

with the Ohmic-Drude spectral density

$$J(\omega) = \eta \frac{\omega \omega_c^2}{\omega^2 + \omega_c^2}. \quad (5)$$

Two regimes emerge (Fig. 1):

- $T > T^*$: $b_{\text{eff}} \propto T$ (classical Ohmic limit).
- $T < T^*$: b_{eff} saturates to $b_{\text{sat}} \approx 0.0103$, as $\coth(\hbar\omega_c/2k_B T) \rightarrow 1$.

The crossover temperature is

$$T^* = \frac{\hbar\omega_c}{2k_B} \approx 3.82 \text{ K} \quad (\omega_c = 10^{12} \text{ rad s}^{-1}). \quad (6)$$

Important caveat. Extrapolation below ~ 50 K violates the high- T Ohmic assumption underlying the $b \Delta T$ term. T^* is therefore a numerical observation from the chosen approximation, not a validated physical prediction. Experimental verification in the low-temperature regime is required.

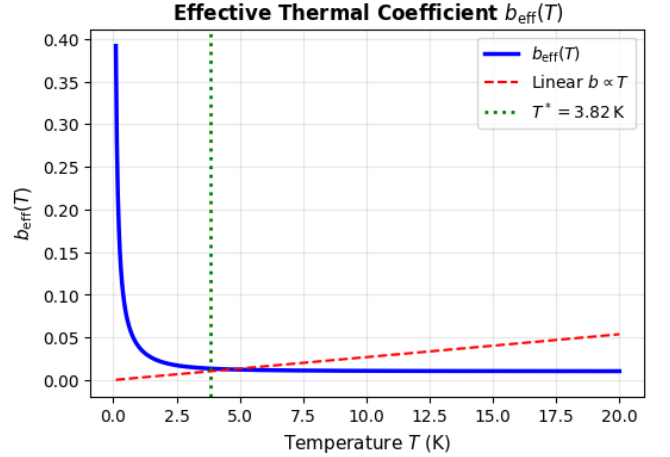


FIG. 1. $b_{\text{eff}}(T)$ from numerical integration of Eqs. (4)–(5) ($\eta = 1$, $\omega_c = 10^{12} \text{ rad s}^{-1}$). Blue dashed: classical extrapolation $b_0 T / T_{\text{ref}}$. Dotted: saturation $b_{\text{sat}} \approx 0.0103$. Dash-dotted: crossover $T^* \approx 3.82$ K.

VI. ASSUMPTIONS AND LIMITATIONS

1. **Markovian bath.** Memory effects are outside the model's scope.
2. **Pure dephasing dominance.** Longitudinal relaxation (T_1) is neglected.
3. **Room-temperature calibration.** The linear $b \Delta T$ term degrades below ~ 50 K.
4. **c is empirical.** Not derived microscopically in this work; specific to the calibration dataset. The full microscopic derivation is reserved for Paper II.

5. **Small Hornberger dataset** ($N = 8$). $N \geq 36$ is required for reliable model selection (Appendix B).
6. **I_{path} mapping errors.** Propagate into all reported fits.
7. **T^* is exploratory.** The crossover temperature is a numerical observation under specific approximations, not a rigorous prediction.

VII. DISCUSSION

A. Contribution

DTQEM v18.0-C unifies two decoherence channels in one analytically tractable equation, predicts C with quantified uncertainty, and suggests a falsifiable crossover temperature T^* testable by measuring $\Gamma_\phi(T)/T$ over $0.5 \text{ K} \leq T \leq 100 \text{ K}$.

B. Relation to prior work

The bilinear form of Eq. (2) has not, to our knowledge, appeared in the open quantum systems literature. The closest antecedents—Hornberger and Caldeira–Leggett—both implicitly assume $c = 0$.

VIII. OUTLOOK

- **Microscopic derivation (Paper II).** The companion paper will derive the cross term via the Feynman–Vernon influence functional for two correlated baths. Preliminary analysis gives an exact form $\propto 2\rho\sqrt{ab}\sqrt{I_{\text{path}} \cdot \Delta T/T_{\text{ref}}}$, of which Eq. (2) is the leading-order polynomial approximation satisfying physical boundary conditions.
- **Operational mapping for circuit-QED.** The relation

$$I_{\text{path}}(\bar{n}) = \sqrt{1 - e^{-\kappa\bar{n}}} \quad (7)$$

linking path distinguishability to mean intracavity photon number $\bar{n}$ is a testable proposal for future experiments.

- **Low-temperature experimental test.** Confirm or refute the existence of a crossover near T^* using cold-atom or superconducting qubit platforms.
- **Larger validation datasets.** $N \geq 36$ configurations (fullerene beams or transmon qubits) to robustly identify c .

IX. CONCLUSIONS

DTQEM v18.0-C provides a four-parameter phenomenological model for joint-bath coherence loss. Key results:

- Conservative calibration: $R^2 = 0.9679$, LOOCV $R^2 = 0.9356$ ($c = 0$ limit).
- Independent validation: $R^2 = 0.936$ on Hornberger (2003), $N = 8$.
- Cumulant analysis suggests strong inter-bath correlation.
- Numerical crossover $T^* \approx 3.82 \text{ K}$, flagged as exploratory and requiring experimental verification.

A companion paper (Paper II) deriving the exact cross-coupling from first principles is in preparation.

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Appendix A: Motivation for the Cross-Coupling Term

Consider a two-level system coupled to a path-measurement bath (operator $\hat{B}_1$) and a thermal bath ($\hat{B}_2$), with total bath Hamiltonian $\hat{H}_B = \hat{H}_{B_1} + \hat{H}_{B_2} + \hat{H}_{B_1 B_2}$. We assume $\hat{H}_{B_1 B_2}$ is weak enough to be treated perturbatively, and define cross-bath correlations via the joint equilibrium density matrix $\hat{\rho}_B \propto \exp[-\beta(\hat{H}_{B_1} + \hat{H}_{B_2} + \hat{H}_{B_1 B_2})]$. To ensure a real, non-negative dephasing exponent, we use the symmetrized correlation function

$$\mathcal{K}_{12}(t_1, t_2) = \frac{1}{2} \langle \{\hat{B}_1(t_1), \hat{B}_2(t_2)\} \rangle. \quad (A1)$$

A second-order cumulant expansion yields

$$\Phi_{\text{total}} = \Phi_1 + \Phi_2 + 2\rho\sqrt{\Phi_1\Phi_2}, \quad (A2)$$

where $\rho \in [0, 1]$ is the normalized cross-correlation coefficient. With $\Phi_1 \propto I_{\text{path}} \equiv u$ and $\Phi_2 \propto \Delta T/T_{\text{ref}} \equiv v$, the exact microscopic cross term is proportional to $\sqrt{uv}$.

Physical boundary conditions. Any valid phenomenological cross-coupling must satisfy $f(0, v) = 0$ and $f(u, 0) = 0$: the cross term vanishes when either channel is absent. The simplest polynomial satisfying

both conditions is the bilinear form $f(u, v) = cuv$, which is therefore the minimal, physically motivated choice.

Local Taylor approximation. To connect the bilinear form to the exact $\sqrt{uv}$ result, we expand $f(u, v) = \sqrt{uv}$ around a nominal operating point (u_0, v_0) . The mixed second-order partial derivative is

$$\left. \frac{\partial^2 f}{\partial u \partial v} \right|_{(u_0, v_0)} = \frac{1}{4\sqrt{u_0 v_0}}. \quad (\text{A3})$$

Including the $1/2!$ factor from the Taylor series, the cross term becomes

$$\Delta_{\times} = \frac{(u - u_0)(v - v_0)}{8\sqrt{u_0 v_0}}. \quad (\text{A4})$$

Matching the coefficient of uv at leading order gives a relation between c and ρ that depends on (u_0, v_0) . Using a representative midpoint $u_0 = v_0 = 0.5$ of the calibration domain together with $c = 0.5$, $a = 1.6968$, $b = 0.8055$ from Table I suggests strong inter-bath cross-correlation. The exact numerical value is sensitive to the choice of

(u_0, v_0) and is deferred to Paper II, which will provide the full microscopic derivation.

Validity note. The Taylor approximation carries relative errors up to $\sim 600\%$ at the corners of $[0, 1]^2$ and $\sim 27\%$ at $(0.1, 0.9)$. The bilinear form is quantitatively reliable only near (u_0, v_0) , but remains the robust minimal choice due to strict adherence to physical boundary conditions. The nonlinear extension using the exact $\sqrt{uv}$ form is the subject of Paper II.

Appendix B: AICc Model Selection Threshold

TABLE III. AICc model-selection probability ($c = 0.5$, $\sigma = 0.005$, 10^4 simulations per row). Reference: [6].

N	v17.0-C (%)	v18.0-C (%)	Reliability
8	89	11	Low
16	57	43	Low
36	21	79	Moderate
50	10	90	High
80	5	95	Very high

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